

外

平成17年度 筑波大学

大学院修士課程 理工学研究科

入学試験問題

大学院博士課程 システム情報工学研究科
(コンピュータサイエンス専攻)

外国語

え い ご
英 語

つぎ 次のページの英文を読んで Problem 1 か Problem 2 のいずれかに答えなさい(両方の問題に答えてはいけない)。Problem 2 は英語で答えよ。

Problem の中の設問 1 と 2 は別々の解答用紙に解答すること。解答用紙には必ず Problem と設問の番号を記入しなさい(例 Problem 2, Question 1)。

Read the paragraphs on the next page and answer either Problem 1 or Problem 2. (Do not answer both.) Answer problem 2 in English.

Answer both questions 1 and 2 of the chosen problem in separate answer sheets, respectively.

Write the number of the chosen problem and the question on each answer sheet. (Ex. Problem 2, Question 1)

For 40 years, scientists have conducted searches for radio signals from an extraterrestrial technology, sent spacecraft to all but one of the planets in our solar system, and greatly expanded our knowledge of the conditions in which living things can survive. ① Assuming our current space program continues, by A.D. 2050 we may finally know whether there is, or ever was, life elsewhere in our solar system. We may find that life is common but technical intelligence is extremely rare or that both are common or rare. A.

Although we talk of searching for extraterrestrial intelligence (SETI), what we are seeking is evidence of extraterrestrial technologies. It might be better to use the acronym SET-T (pronounced the same) to acknowledge this. To date, we have concentrated on a very specific technology—radio transmissions at wavelengths with weak natural backgrounds and little absorption. ② No one has yet found any verified signs of a distant technology. But the null result may have more to do with limitations in range and sensitivity than with actual lack of civilizations. The most distant star probed directly is still less than 1 percent of the distance across our galaxy.

SETI, like all of radio astronomy, now faces a crisis. Humanity's voracious appetite for technologies that utilize the radio spectrum is rapidly obscuring the natural window with curtains of radio-frequency interference. This trend might eventually force us to take our search to the far side of the moon, the one place in the solar system that never has Earth in its sky. B. International agreements have already established a "shielded zone" on the moon. Plans for the next few decades of SETI also envisage the construction of a variety of ground-based instruments that offer greater sensitivity, frequency coverage and observing time.

If by A.D. 2050 we have found no evidence of an extraterrestrial technology, it may be because technical intelligence almost never evolves, or because technical civilizations rapidly bring about their own destruction, or because we have not yet conducted an adequate search using the right strategy. C. By then we may begin considering the active transmission of a signal for someone else to find, at which point we will have to tackle the difficult questions of who will speak for Earth and what they will say.

(Scientific American, December 1999, J. C. Tarter and C. F. Chyba "Is there life elsewhere in the Universe?" より)

ちゅう 注 extraterrestrial: 地球外 ちきゅうがい acronym: 頭字語 (略語) とうじご りゃくご radio astronomy: 電波天文学 でんぱ てんもんがく
 voracious appetite: 飽くなき欲求 あ よつきゅう obscure: 覆い隠す おお かく radio-frequency interference: 電波干渉 でんぱ かんしょう
 envisage: 思い描く おも えが frequency coverage: 周波数帯域 しゅうはすうたいいき evolve: 進化する しんか
 speak for: ~の代弁者となる だいべんしゃ

Problem 1

Question 1. 下線部 ①と②を和訳せよ。

Translate the underlined two sentences, ① and ②, into Japanese.

Question 2. A と B と C には以下の日本語に相当する文章が入る。

以下の文章 A, B, C を英文にせよ。

English sentences corresponding to the following Japanese sentences, A, B, and C, are inserted into A, B, and C in the paragraphs, respectively.

Translate the following three sentences, A, B, and C, into English.

A. 少なくとも、エウロパ (Europa) や火星のような惑星系 (planetary system) に対する組織的な探査がおこなわれているであろう。

B. 月の反対側は地球からの電波干渉の効果がほとんどなく、そのため SETI に対して最も適した場所の1つである。

C. たとえ地球外知性からの信号を捕捉できなくても、SETI によって得られる科学技術の知識は無駄ではない。

Problem 2

Answer the following questions in English.

Question 1.

- (1) Explain the reason why the acronym SET-T is better than the SETI.
- (2) Explain the underlined part ② in detail.

Question 2.

- (1) Explain the reason why SETI now faces a crisis.
- (2) What does no evidence of an extraterrestrial technology by A.D. 2050 mean?
Explain it in detail.
- (3) What are the difficult questions in the last sentence? Explain that in detail.